Field Extensions, Degree, Minimal Polynomials and Irreducibility

From the class book: Chapter 21, Sections 21.1 to 21.3; Chapter 18, Sections 18.2 to 18.7

1. Key definitions

Extension and degree.

- If F is a subfield of E, then E/F is an extension and E is a vector space over F; its dimension $[E : F]$ is the degree.
- The extension is finite if $[E : F] < \infty$.

Algebraic, transcendental.

- $\alpha \in E$ is algebraic over F if $f(\alpha) = 0$ for some non-zero $f \in F[x]$, otherwise transcendental.
- E/F is algebraic if every element of E is algebraic over F.

Minimal polynomial.

- For α algebraic over F, $m_\alpha(x)$ is the unique monic polynomial of least degree in $F[x]$ with root α.
- It is irreducible over F, and $f(\alpha) = 0 \iff m_\alpha \mid f$.

Irreducible polynomial.

- A non-constant $f \in F[x]$ is irreducible over the field F if $f = gh$ forces g or h to be constant.
- Over a domain D, f is irreducible in $D[x]$ if it is a non-zero non-unit and any factorisation has a unit factor.
- So $2x + 2$ is irreducible in $\mathbb{Q}[x]$ but reducible in $\mathbb{Z}[x]$.

Content, primitive. The content of $f \in \mathbb{Z}[x]$ is the gcd of its coefficients; f is primitive if its content is 1.

Cyclotomic polynomial.

- $\Phi_n(x) = \prod (x - \zeta)$ over the primitive n-th roots of unity $\zeta = e^{2\pi i k/n}$, $\gcd(k, n) = 1$.
- It has integer coefficients, degree $\varphi(n)$, and $x^n - 1 = \prod_{d \mid n} \Phi_d(x)$.

2. Main results on extensions

Theorem (Tower law)

If $F \le K \le E$ then $[E : F] = [E : K]\,[K : F]$; products $\alpha_i \beta_j$ of bases of K/F and E/K form a basis of E/F.

Corollaries

- (i) If $[E:F]$ is prime there is no field strictly between F and E.
- (ii) For $\alpha \in E$, $[F(\alpha):F]$ divides $[E:F]$; e.g. a field of degree 2^k over $\mathbb{Q}$ cannot contain $\sqrt[3]{2}$, and a real cubic field cannot contain $\sqrt{2}$.
- (iii) If $[F(\alpha):F]=m$, $[F(\beta):F]=n$, then $[F(\alpha,\beta):F]$ is divisible by m and n and is at most mn; it equals mn when $\gcd(m,n)=1$.

Theorem (Simple algebraic extension)

- If α is algebraic over F with minimal polynomial p of degree n, then $F(\alpha)=F[\alpha]\cong F[x]/\langle p\rangle$, $[F(\alpha):F]=n$, with basis $1,\alpha,\dots,\alpha^{n-1}$.
- If α is transcendental, $F(\alpha)\cong F(x)$ and $[F(\alpha):F]=\infty$.

Theorem (Finite and algebraic)

- Every finite extension is algebraic; the converse fails ($\overline{\mathbb{Q}}$, the algebraic numbers, has $[\overline{\mathbb{Q}}:\mathbb{Q}]=\infty$ since it contains $\sqrt[n]{2}$ for all n).
- Sums, products and quotients of algebraic elements are algebraic, so the algebraic elements form a subfield.
- An algebraic extension of an algebraic extension is algebraic.

Transcendental facts.

- e and π are transcendental over $\mathbb{Q}$; the algebraic numbers are countable, so $[\mathbb{R}:\mathbb{Q}]=\infty$ and almost all reals are transcendental.
- If α is transcendental and $\beta\neq 0$ algebraic, then $\alpha+\beta$, $\alpha\beta$, $\sqrt{\alpha}$ are transcendental. "Algebraic" depends on the base: π is algebraic over $\mathbb{R}$ and over $\mathbb{Q}(\pi)$.
- Numbers such as $\cos\frac{2\pi}{n}$, $\sin(m^\circ)$, $e^{2\pi i/n}$ are algebraic over $\mathbb{Q}$.

3. Main results on irreducibility

Theorem (Degree 2 and 3)

- A polynomial of degree 2 or 3 over a field is reducible iff it has a root in the field.
- False for degree 4: $(x^2+1)^2$ and $x^4+4=(x^2-2x+2)(x^2+2x+2)$ have no rational root.

Theorem (Rational root test)

- If p/q in lowest terms is a root of $a_nx^n+\cdots+a_0\in\mathbb{Z}[x]$, then $p\mid a_0$ and $q\mid a_n$.
- A rational root of a monic integer polynomial is an integer dividing a_0.

Theorem (Gauss's Lemma)

- A product of primitive polynomials is primitive.
- A polynomial in $\mathbb{Z}[x]$ that factors into lower-degree factors in $\mathbb{Q}[x]$ already does so in $\mathbb{Z}[x]$.
- A primitive $f \in \mathbb{Z}[x]$ is irreducible in $\mathbb{Z}[x]$ iff it is irreducible in $\mathbb{Q}[x]$.
- (Same for $D[x]$ and its fraction field, D a UFD.)

Theorem (Mod p test)

- Let $f \in \mathbb{Z}[x]$, $\deg f = n$, and p a prime not dividing the leading coefficient.
- If $\bar{f} \in \mathbb{Z}_p[x]$ is irreducible, then f is irreducible over $\mathbb{Q}$.
- The converse fails: $x^4 + 1$ is irreducible over $\mathbb{Q}$ but reducible mod every prime.

Theorem (Eisenstein's criterion)

- If $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ and a prime p satisfies $p \nmid a_n$, $p \mid a_i$ $(i < n)$, $p^2 \nmid a_0$, then f is irreducible over $\mathbb{Q}$.
- Irreducibility is preserved by $f(x) \mapsto f(x+a)$, $f(cx)$, $cf(x)$ $(c \neq 0)$, so Eisenstein may be applied after a shift.

Theorem (Cyclotomic polynomials)

- $\Phi_p(x) = x^{p-1} + \cdots + x + 1$ is irreducible over $\mathbb{Q}$ (Eisenstein for $\Phi_p(x+1)$); in fact every Φ_n is irreducible over $\mathbb{Q}$.
- Hence $x^n - 1$ has exactly $\tau(n)$ irreducible factors over $\mathbb{Q}$.
- Also $\Phi_{p^k}(x) = \Phi_p(x^{p^{k-1}})$ and $\Phi_{2n}(x) = \Phi_n(-x)$ for odd $n > 1$.

Theorem (Irreducibles over finite fields)

The number of monic irreducible polynomials of degree n over $\mathbb{F}_q$ is

$$N_q(n) = \frac{1}{n} \sum_{d \mid n} \mu(d)\, q^{n/d}, \qquad N_q(2) = \frac{q^2 - q}{2},\ N_q(3) = \frac{q^3 - q}{3},\ N_q(4) = \frac{q^4 - q^2}{4}.$$

Over $\mathbb{F}_q$, $x^{q^n} - x$ is the product of all monic irreducibles of degree dividing n, so $q^n = \sum_{d \mid n} d N_q(d)$.

4. Standard tables

Element over $\mathbb{Q}$	Minimal polynomial	Degree
$\sqrt{2},\ i,\ \omega$	$x^2-2,\ x^2+1,\ x^2+x+1$	2
$\sqrt[n]{p}$, p prime	x^n-p	n
$\sqrt{2}+\sqrt{3}$	x^4-10x^2+1	4
$\sqrt{1+\sqrt{2}}$	x^4-2x^2-1	4
$\sqrt{3}+i$	x^4-4x^2+16	4
$\sqrt[3]{2}+\sqrt[3]{4}$	x^3-6x-6	3
$\sqrt{2}+\sqrt[3]{2}$	a sextic	6
$e^{2\pi i/n}$	$\Phi_n(x)$	$\varphi(n)$
$\cos(2\pi/n),\ n\geq 3$	(real subfield of $\mathbb{Q}(\zeta_n)$)	$\varphi(n)/2$
$2\cos(2\pi/5)$	x^2+x-1	2
$\mathbb{Q}(\sqrt{p_1},\dots,\sqrt{p_n})$, distinct primes	field degree	2^n

n	$\Phi_n(x)$	n	$\Phi_n(x)$
1	$x-1$	8	x^4+1
2	$x+1$	9	x^6+x^3+1
3	x^2+x+1	10	$x^4-x^3+x^2-x+1$
4	x^2+1	12	x^4-x^2+1
5	$x^4+x^3+x^2+x+1$	16	x^8+1
6	x^2-x+1	18	x^6-x^3+1

$N_q(n)$	$n=1$	2	3	4	5	6
$q=2$	2	1	2	3	6	9
$q=3$	3	3	8	18	48	116
$q=5$	5	10	40	150	624	2580

- Over $\mathbb{F}_2$: degree 2: x^2+x+1; degree 3: $x^3+x+1,\ x^3+x^2+1$; degree 4: $x^4+x+1,\ x^4+x^3+1,\ x^4+x^3+x^2+x+1$.
- Over $\mathbb{Z}_p$, x^2+1 is irreducible iff $p\equiv 3\pmod 4$; of the p^2 monic quadratics, $p(p-1)/2$ are irreducible.
- Irreducibles over $\mathbb{C}$: linear only; over $\mathbb{R}$: linear and quadratics with $b^2-4ac<0$.

5. Exam shortcuts and standard counter-examples

Exam shortcut.

- Order of attack over $\mathbb{Q}$: rational root test (degree ≤ 3); Eisenstein, if needed after $x \mapsto x \pm 1$; reduce mod 2 or 3; for a rootless quartic, rule out a product of two quadratics.
- Over $\mathbb{Z}_p$: $f(0) = 0$ gives root 0, coefficient sum $\equiv 0$ gives root 1.
- A quartic over $\mathbb{F}_2$ is irreducible iff it has no root and is not $(x^2 + x + 1)^2 = x^4 + x^2 + 1$.

Exam shortcut.

- To find $[F(\alpha) : F]$: find a polynomial killed by α, prove it irreducible; its degree is the answer.
- To show a degree doubles on adjoining i, note the smaller field lies in $\mathbb{R}$.
- $\mathbb{Q}(\sqrt{a}) = \mathbb{Q}(\sqrt{b})$ iff a/b is a rational square.

Standard counter-examples.

- No rational root yet reducible: $x^4 + 4$, $(x^2 + 1)^2$.
- Irreducible over $\mathbb{Q}$, reducible mod every p: $x^4 + 1$.
- Eisenstein is not necessary: $x^2 + 1$ (use $x \mapsto x + 1$); $x^2 + x + 2$ (no prime works, but negative discriminant).
- Mod p test invalid when p divides the leading coefficient.
- $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ has degree 4, not 2. $\cos \frac{2\pi}{n}$ has degree $\varphi(n)/2$, not $\varphi(n)$.
- $\mathbb{Q}(\sqrt{p}) \ncong \mathbb{Q}(\sqrt{q})$ as fields for distinct primes, though isomorphic as $\mathbb{Q}$-vector spaces.

6. Solved examples

Example 1. Find the minimal polynomial of $\alpha = \sqrt{2} + \sqrt{3}$ over $\mathbb{Q}$ and over $\mathbb{Q}(\sqrt{2})$, and find $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})]$.

Solution

- $\alpha^2 = 5 + 2\sqrt{6}$, so $(\alpha^2 - 5)^2 = 24$: $\alpha^4 - 10\alpha^2 + 1 = 0$.
- Now $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$ (if $\sqrt{3} = a + b\sqrt{2}$, squaring forces $ab = 0$, impossible), so $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$.
- Since $\alpha^{-1} = \sqrt{3} - \sqrt{2}$, both $\sqrt{2} = (\alpha - \alpha^{-1})/2$ and $\sqrt{3} = (\alpha + \alpha^{-1})/2$ lie in $\mathbb{Q}(\alpha)$, so $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, $\deg \alpha = 4$ and $x^4 - 10x^2 + 1$ is the minimal polynomial.
- Over $\mathbb{Q}(\sqrt{2})$: $(\alpha - \sqrt{2})^2 = 3$ gives $x^2 - 2\sqrt{2}\,x - 1$, and the degree is $4/2 = 2$.

Example 2. Find $[\mathbb{Q}(\sqrt{2}, \sqrt[3]{2}) : \mathbb{Q}]$, $[\mathbb{Q}(\sqrt[4]{2}, i) : \mathbb{Q}]$, $[\mathbb{Q}(\sqrt{2}, \sqrt{8}) : \mathbb{Q}]$ and $[\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}(\omega)]$.

Solution

- (a) Degrees 2 and 3 are coprime: 6.
- (b) $x^4 - 2$ is Eisenstein at 2, so $[\mathbb{Q}(\sqrt[4]{2}) : \mathbb{Q}] = 4$; $i \notin \mathbb{Q}(\sqrt[4]{2}) \subset \mathbb{R}$, so the total is 8.
- (c) $\sqrt{8} = 2\sqrt{2}$: degree 2.
- (d) $[\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}] = 6$ (divisible by 2 and 3, at most 6), so over $\mathbb{Q}(\omega)$ the degree is $6/2 = 3$.

Example 3. Prove irreducible over $\mathbb{Q}$: (a) $x^4 + 1$ (b) $x^3 + x^2 - 2x - 1$ (c) $7x^4 + 3x^3 + 2x + 5$ (d) $x^5 - 6x^3 + 9x + 12$.

Solution

- (a) $f(x+1) = x^4 + 4x^3 + 6x^2 + 4x + 2$, Eisenstein at 2.
- (b) Mod 2 it is $x^3 + x^2 + 1$, with no root in $\mathbb{Z}_2$, hence irreducible (degree 3); alternatively $f(x+2) = x^3 + 7x^2 + 14x + 7$ is Eisenstein at 7.
- (c) 2 does not divide 7; mod 2 it is $x^4 + x^3 + 1$: no root (values 1, 1) and it is not $x^4 + x^2 + 1$, so irreducible over $\mathbb{Z}_2$, hence over $\mathbb{Q}$.
- (d) Eisenstein at 3 ($9 \nmid 12$).

Example 4. Find the minimal polynomial and degree of $\beta = \sqrt[3]{2} + \sqrt[3]{4}$ over $\mathbb{Q}$.

Solution

- Let $a = \sqrt[3]{2}$, so $\beta = a + a^2$.
- Then $\beta^3 = a^3 + 3a^4 + 3a^5 + a^6 = 2 + 6a + 6a^2 + 4 = 6 + 6\beta$.
- So β is a root of $x^3 - 6x - 6$, Eisenstein at 2 ($4 \nmid 6$).
- Degree 3, and $\mathbb{Q}(\beta) = \mathbb{Q}(\sqrt[3]{2})$ since $\mathbb{Q}(\beta) \subseteq \mathbb{Q}(a)$ and both have degree 3.

Example 5. (a) Find the degree of $\zeta_{12} = e^{2\pi i/12}$ and of $\cos\frac{2\pi}{7}$ over $\mathbb{Q}$. (b) How many irreducible factors does $x^{12} - 1$ have over $\mathbb{Q}$?

Solution

- (a) $\deg \zeta_{12} = \varphi(12) = 4$, minimal polynomial $\Phi_{12} = x^4 - x^2 + 1$.
- $\deg \cos\frac{2\pi}{7} = \varphi(7)/2 = 3$; in fact $2\cos\frac{2\pi}{7}$ is a root of $x^3 + x^2 - 2x - 1$ (Example 3(b)).
- (b) $\tau(12) = 6$: $\Phi_1, \Phi_2, \Phi_3, \Phi_4, \Phi_6, \Phi_{12}$, of degrees $1, 1, 2, 2, 2, 4$ (sum 12).

Example 6. (a) How many monic irreducible polynomials of degree 4 are there over $\mathbb{F}_3$? (b) How many monic quadratics over $\mathbb{Z}_5$ are irreducible? (c) Is x^2+1 irreducible over $\mathbb{Z}_7$ and over $\mathbb{Z}_{13}$?

Solution

- (a) $N_3(4) = (3^4 - 3^2)/4 = 72/4 = 18$.
- Check: $\sum_{d\mid 4} d N_3(d) = 3 + 2\cdot 3 + 4\cdot 18 = 81$.
- (b) $5 \cdot 4/2 = 10$ (out of 25).
- (c) $7 \equiv 3 \pmod 4$: irreducible over $\mathbb{Z}_7$.
- $13 \equiv 1 \pmod 4$: reducible, $x^2+1 = (x-5)(x+5)$ since $5^2 = 25 \equiv -1 \pmod{13}$.